

Fibonacci Polynomials & *Lights Out*

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Abstract

Lights Out is a game on a simple graph $G = (V, E)$ where clicking a vertex toggles the on/off state of it and its neighbors. Identifying subsets of V with their indicator vectors over $\mathbb{F}_2$, let $\Phi_G : \mathbb{F}_2^V \rightarrow \mathbb{F}_2^V$ be the function which maps the set of vertices clicked to the set of vertices that change state. For the $n \times n$ grid, define $d(n) = \dim(\ker \Phi_G)$. We recover the known fact that $d(n)$ is always even and prove infinitely many even values are also impossible. We provide formulas and bounds for $d(n)$, including a complete characterization when $n = 2^a p^b - 1$ for $a, b \geq 0$ and p not a Wieferich prime.

1 Fibonacci Polynomials

We begin by introducing Fibonacci polynomials. We first develop some basic properties of these polynomials that will be useful in relating them to the Lights Out problem.

Definition 1.1 (Fibonacci Polynomial). *Let $F_n(x)$ be the polynomial in the ring $\mathbb{F}_2[x]$ defined recursively by*

$$F_n(x) = \begin{cases} 0 & n = 0 \\ 1 & n = 1 \\ xF_{n-1}(x) + F_{n-2}(x) & \text{otherwise} \end{cases}.$$

These polynomials are called "Fibonacci" because $F_n(1)$ is the n th Fibonacci number when evaluated over $\mathbb{Z}[x]$. Evaluated over $\mathbb{F}_2$, $F_n(1)$ is the n th Fibonacci number modulo 2.

In $\mathbb{F}_2[x]$, the coefficients of $F_n(x)$ are either 1 or 0, so we can visualize them by coloring squares black or white to represent the coefficients. For example,

$$F_6(x) = 1x^5 + 0x^4 + 0x^3 + 0x^2 + 1x + 0 = \blacksquare \square \square \square \blacksquare \square.$$

Plotting $F_1(x)$ through $F_{128}(x)$ in the same way, aligning terms of the same degree, we see a Sierpinski triangle.

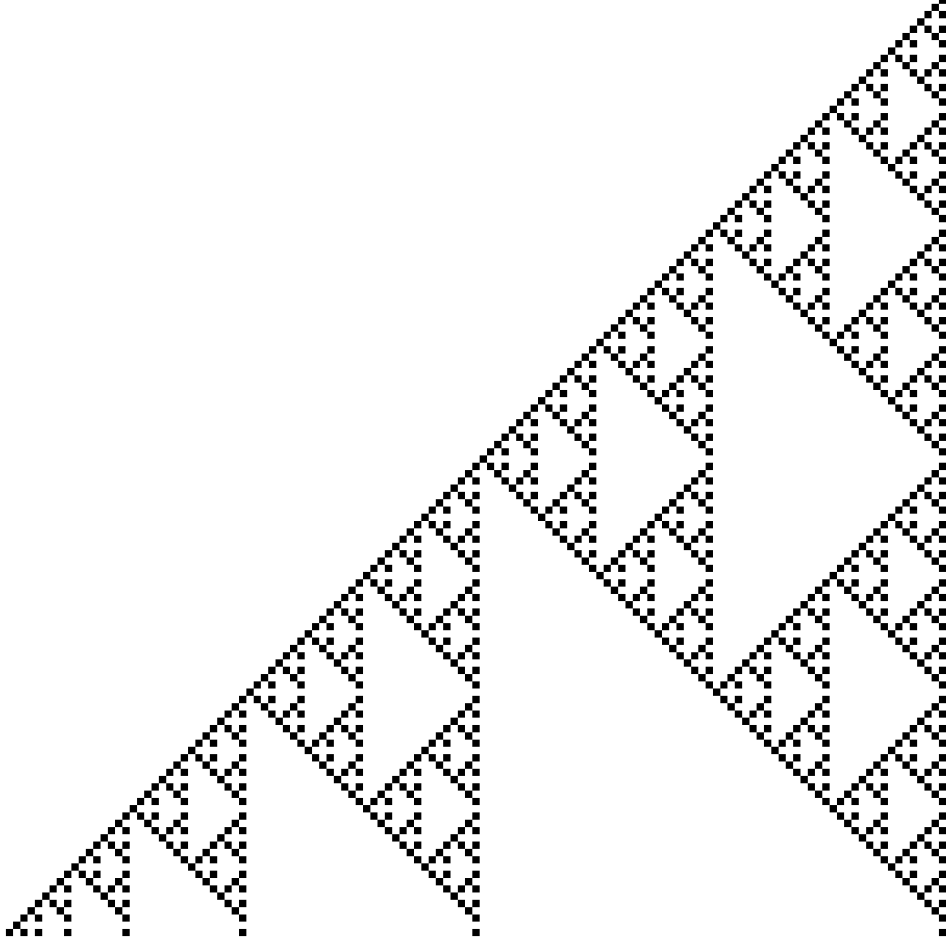


Figure 1: $F_1(x)$ through $F_{128}(x)$, right-aligned.

The Sierpinski arises from the parity of the binomial coefficients in Pascal's triangle, giving the formula

$$F_n(x) = \sum_{i=0}^{n-1} \binom{n+i}{2i+1} x^i \pmod{2}.$$

Sutner also showed the above construction and formula, along with the following relationship between rectangular grids and Fibonacci Polynomials [Sut00, Theorem 5.1].

Lemma 1.2 (Sutner). *Let $G = (V, E)$ be an $n \times m$ grid graph. Then*

$$\dim(\ker \Phi_G) = \deg(\gcd(F_{n+1}(x), F_{m+1}(x+1))) = \deg(\gcd(F_{n+1}(x+1), F_{m+1}(x))).$$

In particular, this lemma allows us to define $d(n)$ in terms of Fibonacci polynomials.

$$d(n) = \deg(\gcd(F_{n+1}(x), F_{n+1}(x+1))).$$

Sutner and others also established several divisibility properties of Fibonacci polynomials [HMP04, Proposition 2.2][Sut00, Proposition 2.1][Knu11].

Lemma 1.3 (Hunziker, Machiavelo, and Park). *A polynomial $\tau(x)$ in $\mathbb{F}_2[x]$ divides both $F_n(x)$ and $F_m(x)$ if and only if it divides $F_{\gcd(m,n)}(x)$. In particular,*

$$\gcd(F_m(x), F_n(x)) = F_{\gcd(m,n)}(x).$$

Corollary 1.4. *The following are true:*

- (i) x divides $F_n(x)$ if and only if $n \equiv 0 \pmod{2}$.
- (ii) $x+1$ divides $F_n(x+1)$ if and only if $n \equiv 0 \pmod{2}$.
- (iii) $x+1$ divides $F_n(x)$ if and only if $n \equiv 0 \pmod{3}$.
- (iv) x divides $F_n(x+1)$ if and only if $n \equiv 0 \pmod{3}$.

Proof. Notice,

- (i) We find that $F_2(x) = x$ is the smallest Fibonacci polynomial divisible by x , so we apply lemma 1.3 to get the desired result.
- (ii) Follows from (i) by substituting $x+1$ for x .
- (iii) We find that $F_3(x) = (x+1)^2$ is the smallest Fibonacci polynomial divisible by $x+1$, so we apply lemma 1.3 to get the desired result.
- (iv) Follows from (iii) by substituting $x+1$ for x .

■

Lemma 1.5 (Hunziker, Machiavelo, and Park). *If n is odd, then $F_n(x)$ is the square of a square-free polynomial.*

Proof. See [HMP04] proposition 2.11 and [Sut00] Theorem 2.1 claim 2. ■

Corollary 1.6. *If n is odd, then every root of $F_n(x)$ has multiplicity 2.*

Proof. Let n be odd and α a root of $F_n(x)$. By 1.5, there exists $\tau_n \in \mathbb{F}_2[x]$ such that

$$F_n(x) = \tau_n(x)^2,$$

and $\tau(x)$ is square-free. Since α is a root of $F_n(x)$, it must also be a root of $\tau_n(x)$ and thus has multiplicity 2. ■

We can now prove some of our own facts about these polynomials and their GCDs, particularly on square grids.

Lemma 1.7. *Let $d(n) = \dim(\ker \Phi_G)$ for G an $n \times n$ grid graph. Then $d(n)$ is always even.*

Proof. Let

$$G_n(x) = \gcd(F_n(x), F_n(x+1)).$$

By lemma 1.2, we know that $d(n) = \deg G_{n+1}(x)$ and $G_n(x) = G_n(x+1)$.

$y = x(x+1)$ is also unchanged by this $x \mapsto x+1$ translation. It is well-known that every polynomial $P(x)$ in $\mathbb{F}_2[x]$ can be uniquely written as

$$P(x) = A(y) + xB(y) \quad A, B \in \mathbb{F}_2[y].$$

We can then determine exactly when P is also unchanged by translation.

$$\begin{aligned} P(x+1) &= P(x) \\ A(y) + (x+1)B(y) &= A(y) + xB(y) \\ B(y) &= 0. \end{aligned}$$

In other words, $P(x)$ is unchanged by translation when P is a polynomial in $x^2 + x$.

Since G_n is also unchanged by translation, there exists $H_n \in \mathbb{F}_2[x]$ such that

$$G_n(x) = H_n(x^2 + x).$$

So,

$$d(n) = \deg G_{n+1}(x) = 2 \deg H_{n+1}(x)$$

and thus $d(n)$ is even. ■

Corollary 1.8. *If n is even, then $d(n)$ is divisible by 4.*

Proof. If n is even, then $n+1$ is odd. Then by 1.5, $F_{n+1}(x)$ and $F_{n+1}(x+1)$ are squares of square-free polynomials $\tau(x)$ and $\tau(x+1)$.

$$\begin{aligned} G_{n+1}(x) &= \gcd(F_{n+1}(x), F_{n+1}(x+1)) \\ &= \gcd(\tau_{n+1}(x)^2, \tau_{n+1}(x+1)^2) \\ &= \gcd(\tau_{n+1}(x), \tau_{n+1}(x+1))^2 \\ \sqrt{G_{n+1}(x)} &= \gcd(\tau_{n+1}(x), \tau_{n+1}(x+1)) \end{aligned}$$

We can see from the right-hand side of the equation that this function is also unaffected by translation. So, by lemma 1.7, there exists $H_{n+1} \in \mathbb{F}_2[x]$ such that

$$\begin{aligned} \sqrt{G_{n+1}(x)} &= H_{n+1}(x^2 + x) \\ G_{n+1}(x) &= H_{n+1}(x^2 + x)^2. \end{aligned}$$

So,

$$d(n) = \deg G_{n+1}(x) = 4 \deg H_{n+1}(x),$$

as desired. ■

Note that the implication does not hold in the other direction: $d(n)$ also be divisible by 4 when n is odd. For example, $d(9) = 8$.

Lemma 1.9. *Let $x = t + t^{-1}$. Then for $t \neq 0$,*

$$xF_n(x) = t^n + t^{-n}.$$

Proof. Define $G_n(t) = t^{n+1} + t^{-(n+1)}$. Then

$$\begin{aligned} xG_{n-1}(t) + G_{n-2}(t) &= (t + t^{-1})(t^n + t^{-n}) + (t^{n-1} + t^{-(n-1)}) \\ &= t^{n+1} + t^{-(n-1)} + t^{n-1} + t^{-(n+1)} + t^{n-1} + t^{-(n-1)} \\ &= t^{n+1} + t^{-(n+1)} \\ &= G_n. \end{aligned}$$

Thus we see that $G_n(t)$ satisfies the same recurrence relation as $F_n(x)$. $xF_n(x)$ also satisfies the same recurrence relation.

$$\begin{aligned} xF_n(x) &= x(xF_{n-1}(x) + F_{n-2}(x)) \\ &= x(xF_{n-1}(x)) + xF_{n-2}(x). \end{aligned}$$

Now we compute $xF_1(x)$ and $xF_2(x)$ in terms of t and G :

$$\begin{aligned} xF_1(x) &= x \\ &= t + t^{-1} \\ &= G_0(t), \end{aligned}$$

and

$$\begin{aligned}
xF_2(x) &= x^2 \\
&= (t + t^{-1})^2 \\
&= t^2 + t^{-2} \\
&= G_1(t).
\end{aligned}$$

Since the first two terms match and they satisfy the same recurrence relation then we've shown by induction that for $t \neq 0$,

$$xF_n(x) = G_{n-1}(t) = t^n + t^{-n}.$$

■

This was also proven in [HMP04], who also prove the following identities.

Lemma 1.10 (Hunziker, Machiavelo, and Park). *Let $n = 2^k b$ where $b, k \in \mathbb{N}$. Then*

$$F_n(x) = x^{2^k-1} F_b(x)^{2^k}.$$

Proof. See [HMP04] lemma 2.6.

■

Lemma 1.11. *The distinct roots of $F_n(x)$ are $\{\zeta + \zeta^{-1} : \zeta^n = 1, \zeta \neq 1\}$, plus 0 when n is even.*

Proof. Notice that $F_n(0) = F_{n-2}(0)$, so

$$F_n(0) = \begin{cases} 1 & n \text{ odd} \\ 0 & n \text{ even} \end{cases}.$$

Thus 0 is a root of $F_n(x)$ whenever n is even.

Let α be a non-zero root of F_n . Choose ζ satisfying

$$\zeta^2 + \alpha\zeta + 1 = 0.$$

Notice that ζ cannot be 0. Dividing by ζ ,

$$\begin{aligned}
\zeta + \alpha + \zeta^{-1} &= 0 \\
\alpha &= \zeta + \zeta^{-1}.
\end{aligned}$$

By, Lemma 1.9,

$$\begin{aligned}
0 &= \alpha F_n(\alpha) \\
&= \zeta^n + \zeta^{-n}
\end{aligned}$$

Since a field has no nonzero nilpotents, we may multiply by ζ^n .

$$\begin{aligned}
0 &= \zeta^{2n} + 1 \\
&= (\zeta^n + 1)^2,
\end{aligned}$$

and

$$\begin{aligned}
0 &= \zeta^n + 1 \\
\zeta^n &= 1.
\end{aligned}$$

Thus the order of ζ divides n . Notice too that $\zeta \neq 1$; otherwise, $\alpha = 1 + 1^{-1} = 0$, which we already showed as not possible.

Conversely, if $\zeta^n = 1$ and $\zeta \neq 1$, then let $\alpha = \zeta + \zeta^{-1}$. Then we see $\alpha \neq 0$ and

$$\alpha F_n(\alpha) = \zeta^n + \zeta^{-n} = 1 + 1^{-1} = 0.$$

Thus, α is a root of F_n .

■

2 Even Dominating Sets

We identify a subset $S \subseteq V$ with its indicator vector $\mathbf{1}_S \in \mathbb{F}_2^V$. Under this identification, symmetric difference of subsets corresponds to addition in $\mathbb{F}_2^V$. The map Φ_G is linear since the state change of each vertex is determined by the parity of the clicks on its closed neighborhood. Consequently, two configurations $V_1, V_2 \subseteq V$ have the same image under Φ_G if and only if $V_1 \triangle V_2 \in \ker \Phi_G$. Thus, configurations producing the same state change form cosets of $\ker \Phi_G$. In particular, whenever an initial configuration is solvable, the set of all solutions is a coset of $\ker \Phi_G$, and hence has 2^d elements, where $d = \dim(\ker \Phi_G)$. $\ker \Phi_G$ also has a natural interpretation in terms of even dominating sets.

Definition 2.1. *An even dominating set of a graph $G = (V, E)$ is a subset $V' \subseteq V$ such that the closed neighborhood $N[v]$ of every vertex contains an even number of vertices from V' .*

Odd dominating sets can be analogously defined. Sutner specifically studied odd dominating sets as the solution to *Lights Out* when all lights are initially on. He showed that such an odd dominating set always exists for any finite graph [Sut89].

Lemma 2.2. *$V' \in \ker \Phi_G$ if and only if V' is an even dominating set of G .*

Proof. Let $V' \in \ker \Phi_G$. Then every vertex is toggled an even number of times when the vertices in V' are clicked. Since each vertex is toggled exactly when the vertices in its closed neighborhood are clicked, for every $v \in V$,

$$|N[v] \cap V'| \text{ is even.}$$

Thus, V' is an even dominating set of G . The converse follows by reversing the same argument. ■

We may now use our graph-theoretic understanding of $\ker \Phi_G$ as even dominating sets to construct even dominating sets on larger grids by tiling those from smaller grids.

Theorem 2.3. *For all $n, k \in \mathbb{N}$,*

$$d(nk - 1) \geq d(n - 1).$$

Proof. Let G_L be an $nk - 1 \times nk - 1$ grid and G_ℓ be an $n - 1 \times n - 1$ grid. Since $nk - 1 = k(n - 1) + (k - 1)$, an G may be broken into a $k \times k$ grid of $n - 1 \times n - 1$ grids, each separated by a horizontal or vertical strip of size 1.

Let Q be an even dominating set of G_ℓ . Let R be Q tiled k times horizontally and vertically, with size 1 separators between each tile, onto G_L where each tile is the reflection across its neighboring separators, as shown below.

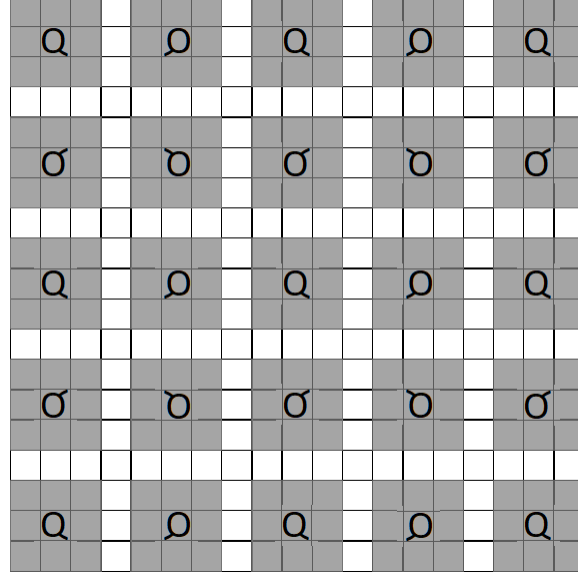


Figure 2: Pattern Q tiled 5 times.

Consider a vertex v in G_L . There are 3 cases.

1. v is within a tile. Then $|N[v] \cap R|$ is even because Q is an even dominating set. Vertices next to tile separators are unaffected by the tiling because the separator vertices are not included in R and thus don't contribute to the parity of $N[v] \cap R$. Thus, $|N[v] \cap R|$ has the same parity as the corresponding closed neighborhood in Q , which is even.
2. v is on a separator between two tiles. Then only two elements in $N[v]$, the vertices within a tile, are also in R . Because neighboring tiles are reflections of each other across separators, these two vertices are either both in R or both not in R . In either case, $|N[v] \cap R|$ is even.
3. v is in the intersection of two separators. Then all of $N[v]$ is in a separator and thus not in R , so $|N[v] \cap R| = 0$, which is even.

Since for every vertex v of G_L , $|N[v] \cap R|$ is even, so R is an even dominating set of G_L . Thus, for every even dominating set of G_ℓ , there exists an even dominating set of G_L obtained through tiling. This tiling function is also injective because we can recover Q by restricting R to just the top left tile. Thus,

$$d(nk - 1) \geq d(n - 1),$$

as desired. ■

Sutner shows a similar result [Sut88]. There is also a simple proof in terms of Fibonacci polynomials.

Proof. Since n divides nk , lemma 1.3 tells us $F_n(x)$ divides $F_{nk}(x)$. Substituting $x + 1$, we also have $F_n(x + 1)$ divides $F_{nk}(x + 1)$. Thus, the first GCD divides the second:

$$\gcd(F_n(x), F_n(x + 1)) \mid \gcd(F_{nk}(x), F_{nk}(x + 1)).$$

So we may write

$$\gcd(F_{nk}(x), F_{nk}(x + 1)) = \gcd(F_n(x), F_n(x + 1))Q(x),$$

for some $Q(x) \in \mathbb{F}_2[x]$. Taking the degrees, we obtain $d(nk - 1) = d(n - 1) + \deg Q(x)$, and the desired inequality follows. ■

Note that although the smaller GCD polynomial divides the larger, it's not the case that $d(n-1)$ divides $d(nk-1)$. For example, if $n = 5$ and $k = 6$, then $d(n-1) = d(4) = 4$, which does not divide $d(nk-1) = d(29) = 10$.

3 Number Theory

We may also understand $d(n)$ and the Fibonacci polynomials with some results from number theory. Many of the results in this section were first discussed by the author in [Boy22b] and [Boy22a]. However, this writing restates several proofs for clarity and to correct some small mistakes in those earlier proofs.

Lemma 3.1 (Ore). *Let a, b, c , and d be integers. Let (a, b) denote $\gcd(a, b)$. Then*

$$(ab, cd) = (a, c)(b, d) \left(\frac{a}{(a, c)}, \frac{d}{(b, d)} \right) \left(\frac{c}{(a, c)}, \frac{b}{(b, d)} \right).$$

Proof. See [O48]. ■

Although Ore's result deals specifically with integers, both the integers and $\mathbb{F}_2[x]$ are Euclidean domains, so the result will still hold for our purposes.

Theorem 3.2. *For all $n \in \mathbb{N}$,*

$$d(2n+1) = 2d(n) + \delta_n, \quad \delta_n = \begin{cases} 2 & 3 \mid n+1 \\ 0 & \text{otherwise} \end{cases}.$$

Proof. Let (a, b) denote $\gcd(a, b)$. Applying lemmas 1.10 and 3.1,

$$\begin{aligned} d(2n+1) &= \deg(F_{2n+2}(x), F_{2n+2}(x+1)) \\ &= \deg(xF_{n+1}(x)^2, (x+1)F_{n+1}(x+1)^2) \\ &= \deg(x, x+1) (F_{n+1}(x)^2, F_{n+1}(x+1)^2) \left(\frac{x+1}{(x, x+1)}, \frac{F_{n+1}(x)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)} \right) \left(\frac{x}{(x, x+1)}, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x)^2, F_{n+1}(x+1)^2)} \right) \\ &= \deg(F_{n+1}(x), F_{n+1}(x+1))^2 \left(x+1, \frac{F_{n+1}(x)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right) \left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right) \\ &= 2d(n) + 2 \deg \left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right) \end{aligned}$$

Thus, we have that

$$\delta_n = 2 \deg \left(x, \frac{F_{n+1}(x+1)^2}{(F_{n+1}(x), F_{n+1}(x+1))^2} \right).$$

So, δ_n is 2 exactly when $F_{n+1}(x)$ can be divided without remainder by $x+1$ more times than $F_{n+1}(x+1)$. When $n+1$ is not divisible by 3, corollary 1.4 tells us that $F_{n+1}(x)$ is not divisible by $x+1$, so $\delta_n = 0$. When $n+1$ is divisible by 3, let $n+1 = b2^k$ for $b, k \geq 0$ and b odd. Then b must be divisible by 3. Applying Lemma 1.10,

$$F_{n+1}(x) = x^{2^k-1} F_b(x)^{2^k} \quad F_{n+1}(x+1) = (x+1)^{2^k-1} F_b(x+1)^{2^k}.$$

Since b is odd, lemma 1.5 tells us $F_b(x)$ is the square of a square-free polynomial. Since b is a multiple of 3, corollary 1.4 tells us that $x+1$ divides $F_b(x)$, but $x+1$ does not divide $F_b(x+1)$. So,

$$F_{n+1}(x) = x^{2^k-1} (x+1)^{2^{k+1}} g(x)^{2^{k+1}} \quad F_{n+1}(x+1) = x^{2^{k+1}} (x+1)^{2^k-1} g(x+1)^{2^{k+1}},$$

for some $g(x) \in \mathbb{F}_2[x]$, where $g(x)$ and $g(x+1)$ are both square-free and divisible by neither x nor $x+1$. So, we see that $F_{n+1}(x)$ can be divided without remainder by $x+1$ more times than $F_{n+1}(x+1)$, so $\delta_n = 2$. ■

This result was first proven by Boyles in [Boy22b] and resolves a conjecture of Sutner's [Sut89].

We can repeatedly apply theorem 3.2 to obtain explicit formulas for $d(n)$.

Corollary 3.3. *Write n as $n+1 = 2^s b$ for $s \geq 0$ and b odd. Then*

$$d(n) = 2^s d(b-1) + \begin{cases} 2(2^s - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases}.$$

Proof. The result is a tautology for even n ($s = 0$ and $n+1 = b$), so we may continue assuming n is odd.

By induction on s . For $s = 0$,

$$\begin{aligned} n+1 &= 2^0 b \\ n &= b-1 \\ d(n) &= d(b-1) \\ &= 2^0 d(b-1) + \begin{cases} 2(2^0 - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases}, \end{aligned}$$

as desired. Assume that our claim holds for $s \geq 0$. Then for $n+1 = 2^{s+1}b$,

$$\begin{aligned} d(n) &= d(2^{s+1}b-1) \\ &= d(2(2^s b-1)+1) \\ &= 2d(2^s b-1) + \begin{cases} 2 & 3 \mid 2^s b \\ 0 & \text{otherwise} \end{cases} \\ &= 2 \left(2^s d(b-1) + \begin{cases} 2(2^s - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases} \right) + \begin{cases} 2 & 3 \mid 2^s b \\ 0 & \text{otherwise} \end{cases} \\ &= 2^{s+1} d(b-1) + \begin{cases} 2(2^{s+1} - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases} + \begin{cases} 2 & 3 \mid b \\ 0 & \text{otherwise} \end{cases} \\ &= 2^{s+1} d(b-1) + \begin{cases} 2(2^{s+1} - 1) & 3 \mid b \\ 0 & \text{otherwise} \end{cases}, \end{aligned}$$

as desired. ■

We can also use theorem 3.2 to narrow down more when $d(n) = 2$.

Corollary 3.4. *If $d(n) = 2$, then $n \equiv -1 \pmod{6}$.*

Proof. Since 2 is not divisible by 4, corollary 1.8 tells us that n must be odd. So, we can write $n = 2^k b - 1$ for b odd and $k \geq 1$. Using corollary 3.3, we can define

$$D_i = d(2^i b - 1) = \begin{cases} 2^i d(b-1) + 2(2^i - 1) & 3 \mid b \\ 2^i d(b-1) & \text{otherwise} \end{cases}.$$

If $3 \nmid b$, then we must have $D_k = 2 = 2^k d(b-1)$. However, corollary 1.8 tells us that $d(b-1)$ is divisible by 4, which cannot be the case.

If $3 \mid b$, then we must have $D_k = 2 = 2^k d(b-1) + 2(2^k - 1)$. If $k \geq 2$, then this expression is greater than 2, which cannot be the case. Thus, we must have $k = 1$ and $D_1 = 2 = 2d(b-1) + 2$, which simplifies to $d(b-1) = 0$. Thus we may write $b = 3j$.

Solving for n , we find

$$\begin{aligned} n &= 2^k b - 1 \\ &= 2b - 1 \\ &= 6j - 1, \end{aligned}$$

so $n \equiv -1 \pmod{6}$, as desired. ■

Definition 3.5. Let P be a non-zero polynomial in $\mathbb{F}_2[x]$. Then

$$\text{rank}(P) = \min \{r > 0 : P(x) \mid F_r(x)\}.$$

Hunziker, Machiavelo, and Park proved that every polynomial divides some $F_r(x)$ and therefore has a finite rank [HMP04] (see Corollary 2.8).

We now show a relationship between divisibility of polynomials and divisibility of their ranks.

Lemma 3.6.

$$P(x) \mid F_m(x) \iff \text{rank}(P) \mid m.$$

Proof. Let $r = \text{rank}(P)$.

Assume that $P(x)$ divides $F_m(x)$. $P(x)$ must also divide $F_r(x)$. So by lemma 1.3,

$$\begin{aligned} P(x) &\mid \gcd(F_r(x), F_m(x)) \\ P(x) &\mid F_{\gcd(r,m)}(x). \end{aligned}$$

Since r is minimal, $\gcd(r, m) = r$, and so $r \mid m$.

Now assume that $r \mid m$. Then $\gcd(r, m) = r$. By lemma 1.3, $F_r(x) \mid F_m(x)$, and so $P(x) \mid F_m(x)$. ■

We can now use this recurrence result to prove certain even values of $d(n)$ are impossible.

Theorem 3.7. The smallest even k such that for all $n \in \mathbb{N}$, $d(n) \neq k$ is $k = 26$.

Proof. Recall from lemma 1.7 that $d(n)$ is always even. Through calculation, we can find that

$$\begin{array}{llllll} d(5) = 2 & d(4) = 4 & d(11) = 6 & d(16) = 8 & d(29) = 10 & d(84) = 12 \\ d(23) = 14 & d(19) = 16 & d(101) = 18 & d(30) = 20 & d(59) = 22 & d(62) = 24. \end{array}$$

We will show that $d(n) = 26$ is impossible.

Assume for the sake of contradiction that there exists $n \in \mathbb{N}$ such that $d(n) = 26$. Since 26 is not divisible by 4, corollary 1.8 says n must be odd. So, we may write $n = 2m + 1$. By theorem 3.2,

$$26 = d(n) = d(2m + 1) = 2d(m) + \begin{cases} 2 & 3 \mid m + 1 \\ 0 & \text{otherwise} \end{cases}.$$

Since lemma 1.7 tells us that $d(m) = 13$ is not possible, we must have $d(m) = 12$ and $m + 1$ is divisible by 3. Let $m + 1 = 2^s b$ for odd b divisible by 3 and $s \geq 0$. By corollary 3.3,

$$d(m) = 2^s d(b-1) + 2(2^s - 1) = 12.$$

The only possible integer solutions here are $d(b-1) = 12$, $s = 0$ or $d(b-1) = 5$, $s = 1$. Lemma 1.7 tells us that $d(b-1) = 5$ is impossible, so the only remaining possibility is that $d(m) = d(b-1) = 12$ and $s = 0$. Thus, m must be even.

Since $m+1$ is odd, lemma 1.5 tells us that there exists some square-free $R_{m+1}(x)$ such that

$$F_{m+1}(x) = R_{m+1}(x)^2.$$

Define $H_{m+1}(x)$ such that

$$H_{m+1}(x) = \gcd(R_{m+1}(x), R_{m+1}(x+1)) \text{ and } \gcd(F_{m+1}(x), F_{m+1}(x+1)) = H_{m+1}(x)^2.$$

Since $R_{m+1}(x)$ is square-free, $H_{m+1}(x)$ is too. Notice too that $H_{m+1}(x) = H_{m+1}(x+1)$. Since $d(m) = 12$, $\deg H_{m+1}(x) = 6$. Lemma 1.7 tells us that for $y = x^2 + x$, there exists a square-free degree 3 polynomial $h_{m+1}(y) \in \mathbb{F}_2[y]$ such that

$$H_{m+1}(x) = h_{m+1}(y).$$

Since $m+1$ is odd, we must also require that $h_{m+1}(0) = 1$. There are only three such polynomials in $\mathbb{F}_2[y]$ that satisfy all these requirements:

1. $C_1(y) = y^3 + 1$
2. $C_2(y) = y^3 + y + 1$
3. $C_3(y) = y^3 + y^2 + 1$.

We will handle these case-by-case.

Calculating the rank of our three candidates for $h_{m+1}(x)$ with respect to x ,

$$\begin{aligned} \text{rank}((x^2 + x)^3 + 1) &= 85 \\ \text{rank}((x^2 + x)^3 + (x^2 + x) + 1) &= 63 \\ \text{rank}((x^2 + x)^3 + (x^2 + x)^2 + 1) &= 63 \end{aligned}$$

If either rank-63 polynomial $C_2(x)$ or $C_3(x)$ is actually $h_{m+1}(x)$, it must also divide $F_{m+1}(x)$ because $h_{m+1} \mid R_{m+1} \mid F_{m+1}$. Thus, by lemma 3.6, 63 would divide $m+1$. However, both polynomials would divide $R_{m+1}(x)$. C_2 and C_3 are coprime and each of degree 6 in x , so $\deg H_{m+1}(x) \geq \deg C_2(x) + \deg C_3(x) = 12$, contradicting our assumption that the degree was 6. Thus neither C_2 nor C_3 can be $h_{m+1}(x)$.

Similarly, if the rank-85 polynomial $C_1(x)$ is actually $h_{m+1}(x)$, then lemma 3.6 tells us 85 divides $m+1$. We also showed earlier that 3 also divides $m+1$. Since $\text{lcm}(85, 3) = 255$ must divide $m+1$.

Let

$$P(x) = x^4 + x^3 + 1 \text{ and } Q(x) = x^4 + x^3 + x^2 + x + 1.$$

P and Q are closed under translation ($P(x+1) = Q(x)$), are coprime to each other, and are coprime with C_1 . The ranks of $P(x)$ and $Q(x)$ are 15 and 17 respectively. By lemma 3.6, since both 15 and 17 divide 255, and 255 divides $m+1$, then both $P(x)$ and $Q(x)$ must divide $H_{m+1}(x)$. Therefore,

$$\deg H_{m+1}(x) \geq \deg C_1(x) + \deg P(x) + \deg Q(x) = 14,$$

which contradicts our assumption that the degree was 6.

We showed there are no suitable polynomials, so we conclude there is no $3 \mid m+1$ such that $d(m) = 12$ and therefore no n such that $d(n) = 26$. ■

Now that we know that one impossible even value of $d(n)$ exists, we can show an entire family of impossible even values also exists.

Corollary 3.8. *Suppose $a \in \mathbb{N}$ is even and an impossible value of $d(n)$. Then $2a+2$ is also an impossible value of $d(n)$.*

Proof. Suppose for the sake of contradiction that there exists n such that $d(n) = 2a + 2$. Since $2a + 2$ is not divisible by 4, then corollary 1.8 tells us that n must be odd, so we may write $n = 2m + 1$. So, by theorem 3.2,

$$2a + 2 = d(n) = d(2m + 1) = 2d(m) + \delta_m \quad \delta_m = \begin{cases} 2 & 3 \mid m + 1 \\ 0 & \text{otherwise} \end{cases}.$$

If $\delta_m = 0$, then $2d(m)$ is divisible by 4, which cannot be the case since we know the value is $2a + 2$. If $\delta_m = 2$, then $d(m) = a$, contradicting the impossibility of a as an output of d . ■

Expanding the recurrence, if a is an impossible value, then $2^t(a + 2) - 2$ is also impossible for all $t \geq 0$. In other words, 26, 54, 110, 222, 446, ... are also all impossible values of $d(n)$.

4 Finite Fields

Although we have already been working in $\mathbb{F}_2$, we will now more directly use results from the study of finite fields.

Lemma 4.1. *Let p be an odd prime. Define $h = \text{ord}_p(2)$ and $c = v_p(2^h - 1)$, where $v_p(a)$ is the p -adic valuation of a . Then for every $j \geq 1$,*

$$\text{ord}_{p^j}(2) = hp^{\max(0, j-c)}.$$

Proof. Suppose $2^r \equiv 1 \pmod{p^j}$. Then h divides r , so we may write $r = ht$. By the Lifting the Exponent Lemma [IR90],

$$\begin{aligned} v_p(2^r - 1) &= v_p(2^{ht} - 1) \\ &= v_p(2^h - 1) + v_p(t) \\ &= c + v_p(t). \end{aligned}$$

So, $p^j \mid 2^{ht} - 1$ exactly when $c + v_p(t) \geq j$. The smallest possible t satisfying this inequality is $t = p^{\max(0, j-c)}$, so

$$\text{ord}_{p^j}(2) = hp^{\max(0, j-c)},$$

as desired. ■

Definition 4.2. *A prime number p is a Wieferich prime when*

$$p^2 \mid 2^{p-1} - 1.$$

Although infinitely many Wieferich primes are conjectured to exist, the only ones known to date are 1093 and 3511 [CDP97], and any larger ones must be greater than 2^{64} [Col26].

We will show an equivalent definition of Wieferich primes.

Lemma 4.3. *An odd prime number p is a Wieferich prime if and only if*

$$p^2 \mid 2^{\text{ord}_p(2)} - 1.$$

Proof. Let $n = \text{ord}_p(2)$. Then $p - 1 = kn$ for some natural number k , and

$$2^{p-1} - 1 = 2^{kn} - 1.$$

So, by the Lifting the Exponent lemma [IR90],

$$v_p(2^{kn} - 1) = v_p(2^n - 1) + v_p(k).$$

However, $k = (p - 1)/n$ is smaller than p and thus not divisible by p , so $v_p(k) = 0$. Thus,

$$v_p(2^{p-1} - 1) = v_p(2^n - 1).$$

So, $p^2 \mid 2^{p-1} - 1$ if and only if $p^2 \mid 2^{\text{ord}_p(2)} - 1$, as desired. ■

Corollary 4.4. *Let p be an odd prime. If p isn't a Wieferich prime, then*

$$\text{ord}_{p^j}(2) = \text{ord}_p(2) p^{j-1}.$$

Proof. Lemma 4.1 tells us that

$$\text{ord}_{p^j}(2) = \text{ord}_p(2) p^{\max(0, j-c)}, \quad c = v_p(2^{\text{ord}_p(2)} - 1).$$

Thus it is sufficient to show that $c = 1$. By definition of $\text{ord}_p(2)$, $c \geq 1$. p is not a Wieferich prime, so by lemma 4.3, for all $t \geq 2$,

$$p^t \nmid 2^{\text{ord}_p(2)} - 1.$$

Thus $c < 2$ and therefore must be exactly 1, as desired. ■

Definition 4.5. *Let b be coprime to n . Then “signed order” of b modulo n is*

$$\rho_b(n) = \min \{r > 0 : b^r \equiv \pm 1 \pmod{n}\}.$$

This is also sometimes called the “index” of ± 1 relative to b , written $\text{ind}_b(\pm 1)$.

Lemma 4.6. *Let p be an odd prime, $j \geq 1$, and $H_j = \text{ord}_{p^j}(2)$. Then*

$$\rho_2(p^j) = \begin{cases} H_j/2 & H_j \text{ is even} \\ H_j & H_j \text{ is odd} \end{cases}.$$

Moreover, when H_j is even,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

Proof. Assume H_j is even. Then by definition of H_j ,

$$\left(2^{H_j/2}\right)^2 \equiv 1 \pmod{p^j}.$$

Thus the only solutions are $2^{H_j/2} \equiv \pm 1 \pmod{p^j}$. However, 1 cannot be a solution; otherwise, H_j would not be minimal. So,

$$2^{H_j/2} \equiv -1 \pmod{p^j}.$$

Notice that if $r \leq H_j/2$, and $2^r \equiv -1 \pmod{p^j}$, then $2^{2r} \equiv 1 \pmod{p^j}$; since H_j is minimal, $H_j \leq 2r$. Thus to satisfy both constraints we must have $H_j/2 = r$.

Now assume H_j is odd. Notice that by definition,

$$H_j = \text{ord}_{p^j}(2) = \min \{h > 0 : 2^h \equiv 1 \pmod{p^j}\},$$

so smaller solutions can only be of the form $2^r \equiv -1 \pmod{p^j}$. Then $2^{2r} \equiv 1 \pmod{p^j}$, which means $H_j \mid 2r$, and since H_j is odd, $H_j \mid r$. Thus $H_j \leq r$, so H_j is minimal, as desired. ■

We can use this notion of a signed order to find the extension degree of extensions of $\mathbb{F}_2$.

Lemma 4.7. *Let $M > 1$ be odd; let ζ have exact order M ; set $\alpha = \zeta + \zeta^{-1}$. Then*

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

Proof. A standard finite field fact says

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min \{r > 0 : \alpha^{2^r} = \alpha\}.$$

By definition of α , $\alpha^{2^r} = \zeta^{2^r} + \zeta^{-2^r}$, so $\alpha^{2^r} = \alpha$ is equivalent to $\zeta^{2^r} + \zeta^{-2^r} = \zeta + \zeta^{-1}$.

For any non-zero $a, b \in \overline{\mathbb{F}_2}$, $a + a^{-1} = b + b^{-1} \iff (a+b)(ab+1) = 0$. So, either $a = b$ or $a = b^{-1}$. Taking $a = \zeta^{2^r}$ and $b = \zeta$, we obtain

$$\zeta^{2^r} = \zeta \quad \text{or} \quad \zeta^{2^r} = \zeta^{-1}.$$

Since ζ has exact order M , we can equivalently write

$$2^r \equiv 1 \pmod{M} \quad \text{or} \quad 2^r \equiv -1 \pmod{M}.$$

So, $\alpha^{2^r} = \alpha \iff 2^r \equiv \pm 1 \pmod{M}$, and

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \min \{r > 0 : \alpha^{2^r} = \alpha\} = \rho_2(M),$$

as desired. ■

Lemma 4.8. *Let a be algebraic over $\mathbb{F}_q$. Then for every $z \in \mathbb{F}_q(a)$,*

$$\text{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(z) = \sum_{r=0}^{[\mathbb{F}_q(a):\mathbb{F}_q]-1} z^{q^r}.$$

Proof. See [LN97]. ■

Corollary 4.9.

$$\text{Tr}_{\mathbb{F}_q(a)/\mathbb{F}_q}(1) = [\mathbb{F}_q(a) : \mathbb{F}_q].$$

Proof. Immediate from evaluating lemma 4.8 at $z = 1$. ■

Lemma 4.10. *Let $M > 1$ be odd, and let ζ have exact order M . Let $\alpha = \zeta + \zeta^{-1}$. Define*

$$G_M = \{\pm 2^r \pmod{M} : r \geq 0\} \subseteq (\mathbb{Z}/M\mathbb{Z})^\times.$$

Then

$$\text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

Proof. Lemma 4.7 showed that

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(M).$$

So, by lemma 4.8,

$$\text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{r=0}^{\rho_2(M)-1} \zeta^{2^r} + \zeta^{-2^r}.$$

By the definition of $\rho_2(M)$, the residues $2^0, -2^0, 2^1, -2^1, \dots, 2^{\rho_2(M)-1}, -2^{\rho_2(M)-1}$ are all distinct modulo M . Otherwise, we would obtain $2^r \equiv \pm 2^s \pmod{M}$, or equivalently, $2^{r-s} \equiv \pm 1 \pmod{M}$ for some $0 \leq s < r < \rho_2(M)$, which contradicts $\rho_2(M)$ being minimal.

The residues are exactly the elements of G_M , so

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u,$$

as desired. ■

Corollary 4.11. *If $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$, then*

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \mu(M) \pmod{2},$$

where μ is the number-theoretic Möbius function.

Proof. From lemma 4.10, we know that

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{u \in G_M} \zeta^u.$$

When $G_M = (\mathbb{Z}/M\mathbb{Z})^\times$, we also have

$$\mathrm{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \sum_{0 \leq u < M, \gcd(u, M)=1} \zeta^u = \mu(M) \pmod{2},$$

by Ramanujan's sum identity of roots of unity, as desired. ■

Now we may combine together many of our partial results to state a more significant result.

Theorem 4.12. *If p is not a Wieferich prime, then for all $k \geq 1$,*

$$d(p^k - 1) = d(p - 1).$$

Proof. Let p be a non-Wieferich prime. The case for $p = 2$ was shown in [Sut89], so we will consider the case when p is odd. For $1 \leq j \leq k$, let

$$R_j = \{\zeta + \zeta^{-1} : \mathrm{ord}(\zeta) = p^j\}.$$

By lemma 1.11,

$$\mathrm{roots} F_{p^k} = \bigcup_{j=1}^k R_j.$$

Furthermore, since p^k is odd, corollary 1.6 and lemma 1.5 tell us that F_{p^k} is the square of a square-free polynomial. In other words, F_{p^k} has $(p^k - 1)/2$ distinct roots. Let $h = \mathrm{ord}_p(2)$. Since p is not Wieferich, corollary 4.4 tells us that

$$\mathrm{ord}_{p^j}(2) = \mathrm{ord}_p(2) p^{j-1},$$

and lemma 4.6 tells us that

$$\rho_2(p^j) = \rho_2(p) p^{j-1}.$$

So, by lemma 4.7, if $\alpha \in R_j$,

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = \rho_2(p) p^{j-1}.$$

Suppose both α and $\alpha + 1$ are roots of F_{p^k} with $\alpha \in R_j$ and $\alpha + 1 \in R_\ell$. Since $\mathbb{F}_2(\alpha) = \mathbb{F}_2(\alpha + 1)$, that is they are the same field with the same degree, α and $\alpha + 1$ must both be members of the same R_j and so $j = \ell$.

Suppose for the sake of contradiction that α and $\alpha + 1$ are both members of R_j for some $j \geq 2$. Set $M = p^j$ and $m = p^{j-1}$ and choose a primitive M th root ζ such that $\alpha = \zeta + \zeta^{-1}$. Since $\alpha + 1$ is also in R_j , there must exist s such that $\gcd(s, p) = 1$ and $\alpha + 1 = \zeta^s + \zeta^{-s}$. Then

$$1 + \alpha + (\alpha + 1) = 1 + \zeta + \zeta^{-1} + \zeta^s + \zeta^{-s} = 0.$$

Equivalently, ζ is a root of

$$P_s(x) = 1 + x + x^{M-1} + x^s + x^{M-s}.$$

Notice that ζ^m is a primitive p th root. Let $f(x)$ be the minimal polynomial of ζ^m over $\mathbb{F}_2$. The degree of $f(x)$ is $\text{ord}_p(2)$. Since $f(\zeta^m) = 0$, ζ is a root of $f(x^m)$. The degree of $f(x^m)$ is $\text{ord}_p(2)m = \text{ord}_{p^j}(2)$. This is also the degree of the minimal polynomial of ζ and so

$$f(x^m) \mid P_s(x).$$

So, every polynomial can be decomposed by exponents modulo m :

$$P_s(x) = \sum_{a=0}^{m-1} x^a P_a(x^m).$$

Since $f(x^m) \mid P_s(x)$, each $P_a(y)$ must be divisible by $f(y)$. However, only exponent 0 among $0, 1, M-1, s, M-s$ is divisible by m , so $P_0(y) = 1$, a contradiction. Instead, we must conclude that no common roots of $F_{p^k}(x)$ and $F_{p^k}(x+1)$ are in R_j for $j \geq 2$.

So the only common roots of $F_{p^k}(x)$ and $F_{p^k}(x+1)$ must come from R_1 and so we must conclude

$$d(p^k - 1) = d(p - 1).$$

■

We can combine this result with our recurrence for $d(2n+1)$.

Corollary 4.13. *Let p be an odd non-Wieferich prime. Then for all $n, m \geq 0$,*

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p-1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

Proof. The $m = 0$ case reduces to what Sutner already showed [Sut89]. For the other cases, we first apply corollary 3.3 to obtain

$$d(2^n p^m - 1) = \begin{cases} 0 & m = 0 \\ 2^n d(p^m - 1) & m \geq 1, p \neq 3 \\ 2^{n+1} - 2 & m \geq 1, p = 3 \end{cases}.$$

We then apply theorem 4.12 to change $d(p^m - 1)$ to $d(p - 1)$.

■

Corollary 4.14. *For all $n, m \geq 1$, the following relationships hold*

1. $d(3^n - 1) = 0$
2. $d(5^n - 1) = 4$
3. $d(7^n - 1) = 0$
4. $d(17^n - 1) = 8$
5. $d(31^n - 1) = 20$
6. $d(127^n - 1) = 56$
7. $d(257^n - 1) = 144$
8. $d(683^n - 1) = 132$

9. $d(8191^n - 1) = 4004$
10. $d(2^n 3^m - 1) = 2^{n+1} - 2$
11. $d(2^n 5^m - 1) = 2^{n+2}$
12. $d(2^n 7^m - 1) = 0$
13. $d(2^n 17^m - 1) = 2^{n+3}$
14. $d(2^n 31^m - 1) = 2^{n+2} \cdot 5$
15. $d(2^n 127^m - 1) = 2^{n+3} \cdot 7$
16. $d(2^n 257^m - 1) = 2^{n+4} \cdot 9$
17. $d(2^n 683^m - 1) = 2^{n+2} \cdot 33$
18. $d(2^n 8191^m - 1) = 2^{n+2} \cdot 7 \cdot 11 \cdot 13$

You may have noticed that for many sizes, we have $d(p-1) = 0$. We can actually classify some p for which this is the case.

Theorem 4.15. *Let $p \equiv 3 \pmod{4}$ be a prime where*

$$\{\pm 2^r \pmod{p} : r \geq 0\} = (\mathbb{Z}/p\mathbb{Z})^\times,$$

or equivalently,

$$\text{ord}_p(2) \in \{p-1, \frac{p-1}{2}\}$$

Then $d(p-1) = 0$.

Proof. Assume for the sake of contradiction that $d(p-1) \neq 0$. Then $F_p(x)$ and $F_p(x+1)$ have a common factor and thus a common root α , so

$$F_p(\alpha) = F_p(\alpha+1) = 0.$$

By lemma 1.11, there must exist non-trivial p th roots of unity ζ and η such that

$$\alpha = \zeta + \zeta^{-1} \quad \alpha + 1 = \eta + \eta^{-1}.$$

We can see that $\mathbb{F}_2(\alpha+1) = \mathbb{F}_2(\alpha)$ (α and $\alpha+1$ can be obtained from each other by adding 1), and so by corollary 4.11,

$$\text{Tr}_{\mathbb{F}_2(\alpha+1)/\mathbb{F}_2}(\alpha+1) = \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) = \mu(p) \pmod{2} = 1.$$

However, by lemmas 4.7 and 4.6,

$$[\mathbb{F}_2(\alpha) : \mathbb{F}_2] = [\mathbb{F}_2(\alpha+1) : \mathbb{F}_2] = \rho_2(p) = \frac{p-1}{2}.$$

Since $p \equiv 3 \pmod{4}$, this value is odd. Applying the linearity of the trace and corollary 4.9,

$$\text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha+1) = \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) + \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(1) = \text{Tr}_{\mathbb{F}_2(\alpha)/\mathbb{F}_2}(\alpha) + 1 = 0.$$

Thus we have reached a contradiction. ■

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